

General Introduction Algorithms

Let's write a program

- ◆ A program is a collection of statements written in a language the computer understands.
- ◆ A computer executes program statements one after another **in sequence** until it reaches the end of the program unless some statement in the program alters the order of execution.
- ◆ Let's write a program to accept two numbers from the user, multiply them together and display the result on the screen.
- ◆ We'll write the program in COBOL.
- ◆ What COBOL program statements will we need to do the job?



Program Statements

- ◆ We need a statement to take in the first number and store it in a named memory location.

❓ **ACCEPT Num1.**

- ◆ We need a statement to take in the second number and store it in a named memory location.

❓ **ACCEPT Num2.**

- ◆ We need a statement to multiply the two numbers together and store the result in a named location.

❓ **MULTIPLY Num1 BY Num2 GIVING Result.**

- ◆ We need a statement to display the result on the screen.

❓ **DISPLAY "Result is = ", Result.**



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.  FragmentA.  
AUTHOR.  Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
0	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
0	5	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

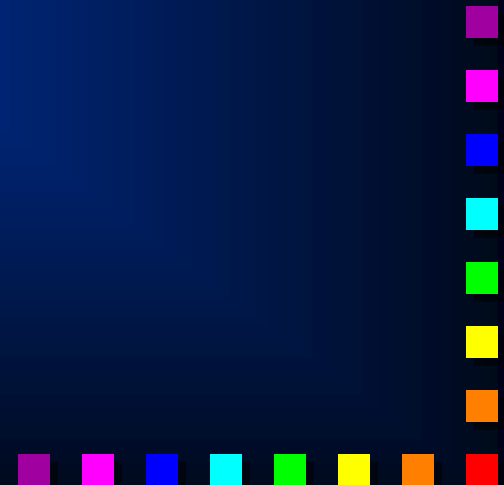
```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA		
Num1	Num2	Result
9	5	0

**Oh No!!
We got the
wrong
answer.
Let's try
again.**




```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    STOP RUN.
```

DATA		
Num1	Num2	Result
0	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	5	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

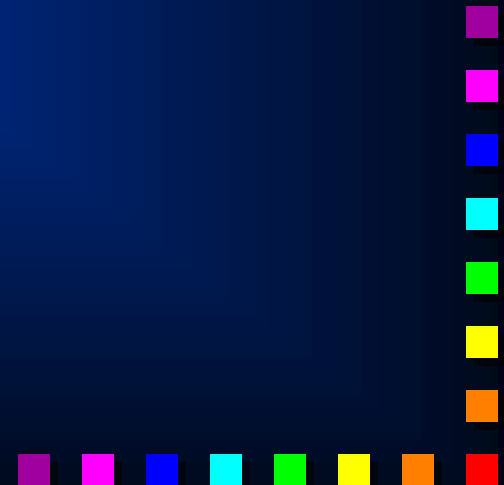
```
    DISPLAY "Result is = ", Result.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    STOP RUN.
```

DATA		
Num1	Num2	Result
9	5	0

**Oh No!!
We got the
wrong answer
again.
Let's have one
more try.**



```
$ SET SOURCEFORMAT"FREE"  
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    DISPLAY "Result is = ", Result.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	5	45

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
0	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	0	00

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result
9	5	00


```
$ SET SOURCEFORMAT"FREE"  
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.
```

```
Calc-Result.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    MULTIPLY Num1 BY Num2 GIVING Result.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA		
Num1	Num2	Result
9	5	45

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.
```

```
PROCEDURE DIVISION.  
Calc-Result.  
    ACCEPT Num1.  
    ACCEPT Num2.  
    MULTIPLY Num1 BY Num2 GIVING Result.  
    DISPLAY "Result is = ", Result.  
    STOP RUN.
```

DATA

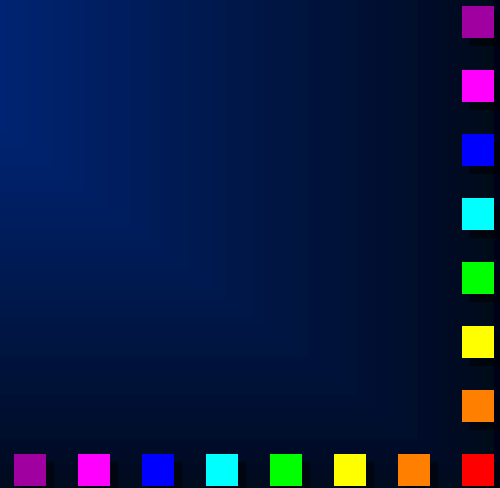
Num1 Num2 Re

Right .. At last!!

**It's not enough to
know what
statements we
need. We must
make sure the
computer executes
them in the right
order.**

Programs are written using three main programming constructs.

- ◆ Sequence.
- ◆ Selection.
- ◆ Iteration.



```
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```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.  
01  Num2          PIC 9  VALUE ZEROS.  
01  Result        PIC 99 VALUE ZEROS.  
01  Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.  
    ACCEPT Num2.  
    ACCEPT Operator  
    IF Operator = "+" THEN  
        ADD Num1, Num2 GIVING Result  
    END-IF  
    IF Operator = "*" THEN  
        MULTIPLY Num1 BY Num2 GIVING Result  
    END-IF.  
    DISPLAY "Result is = ", Result.  
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
0	0	00	_

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Selection-Program.  
AUTHOR. Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01 Num1          PIC 9  VALUE ZEROS.  
01 Num2          PIC 9  VALUE ZEROS.  
01 Result        PIC 99 VALUE ZEROS.  
01 Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
ACCEPT Num1.
```

```
ACCEPT Num2.
```

```
ACCEPT Operator
```

```
IF Operator = "+" THEN
```

```
ADD Num1, Num2 GIVING Result
```

```
END-IF
```

```
IF Operator = "*" THEN
```

```
MULTIPLY Num1 BY Num2 GIVING Result
```

```
END-IF.
```

```
DISPLAY "Result is = ", Result.
```

```
STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	0	00	_

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.
01  Num2          PIC 9  VALUE ZEROS.
01  Result        PIC 99 VALUE ZEROS.
01  Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    ACCEPT Operator
```

```
    IF Operator = "+" THEN
```

```
        ADD Num1, Num2 GIVING Result
```

```
    END-IF
```

```
    IF Operator = "*" THEN
```

```
        MULTIPLY Num1 BY Num2 GIVING Result
```

```
    END-IF.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	3	00	_

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

DATA DIVISION.

WORKING-STORAGE SECTION.

```
01  Num1          PIC 9  VALUE ZEROS.
01  Num2          PIC 9  VALUE ZEROS.
01  Result        PIC 99 VALUE ZEROS.
01  Operator      PIC X  VALUE SPACE.
```

PROCEDURE DIVISION.

Calculator.

ACCEPT Num1.

ACCEPT Num2.

ACCEPT Operator

IF Operator = "+" THEN

ADD Num1, Num2 GIVING Result

END-IF

IF Operator = "*" THEN

MULTIPLY Num1 BY Num2 GIVING Result

END-IF.

DISPLAY "Result is = ", Result.

STOP RUN.

DATA

Num1	Num2	Result	Operator
7	3	00	+

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.
01  Num2          PIC 9  VALUE ZEROS.
01  Result        PIC 99 VALUE ZEROS.
01  Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    ACCEPT Operator
```

```
    IF Operator = "+" THEN
```

```
        ADD Num1, Num2 GIVING Result
```

```
    END-IF
```

```
    IF Operator = "*" THEN
```

```
        MULTIPLY Num1 BY Num2 GIVING Result
```

```
    END-IF.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	3	00	+


```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.
01  Num2          PIC 9  VALUE ZEROS.
01  Result        PIC 99 VALUE ZEROS.
01  Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    ACCEPT Operator
```

```
    IF Operator = "+" THEN
```

```
        ADD Num1, Num2 GIVING Result
```

```
    END-IF
```

```
    IF Operator = "*" THEN
```

```
        MULTIPLY Num1 BY Num2 GIVING Result
```

```
    END-IF.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	3	10	+

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE ZEROS.
01  Num2          PIC 9  VALUE ZEROS.
01  Result        PIC 99 VALUE ZEROS.
01  Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    ACCEPT Operator
```

```
    IF Operator = "+" THEN
```

```
        ADD Num1, Num2 GIVING Result
```

```
    END-IF
```

```
    IF Operator = "*" THEN
```

```
        MULTIPLY Num1 BY Num2 GIVING Result
```

```
    END-IF.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	3	10	+

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Selection-Program.  
AUTHOR. Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01 Num1          PIC 9  VALUE ZEROS.  
01 Num2          PIC 9  VALUE ZEROS.  
01 Result        PIC 99 VALUE ZEROS.  
01 Operator      PIC X  VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

```
    ACCEPT Operator
```

```
    IF Operator = "+" THEN
```

```
        ADD Num1, Num2 GIVING Result
```

```
    END-IF
```

```
    IF Operator = "*" THEN
```

```
        MULTIPLY Num1 BY Num2 GIVING Result
```

```
    END-IF.
```

```
    DISPLAY "Result is = ", Result.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
7	3	10	+

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Iteration-Program.  
AUTHOR. Michael Coughlan.
```

DATA DIVISION.

WORKING-STORAGE SECTION.

```
01 Num1          PIC 9  VALUE ZEROS.  
01 Num2          PIC 9  VALUE ZEROS.  
01 Result        PIC 99 VALUE ZEROS.  
01 Operator      PIC X  VALUE SPACE.
```

PROCEDURE DIVISION.

Calculator.

```
    PERFORM 5 TIMES  
        ACCEPT Num1  
        ACCEPT Num2  
        ACCEPT Operator  
        IF Operator = "+" THEN  
            ADD Num1, Num2 GIVING Result  
        END-IF  
        IF Operator = "*" THEN  
            MULTIPLY Num1 BY Num2 GIVING Result  
        END-IF  
        DISPLAY "Result is = ", Result  
    END-PERFORM.  
STOP RUN.
```

DATA

Num1	Num2	Result	Operator
0	0	00	_

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Iteration-Program.  
AUTHOR. Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01 Num1 PIC 9 VALUE ZEROS.  
01 Num2 PIC 9 VALUE ZEROS.  
01 Result PIC 99 VALUE ZEROS.  
01 Operator PIC X VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
PERFORM 5 TIMES
```

```
ACCEPT Num1
```

```
ACCEPT Num2
```

```
ACCEPT Operator
```

```
IF Operator = "+" THEN
```

```
ADD Num1, Num2 GIVING Result
```

```
END-IF
```

```
IF Operator = "*" THEN
```

```
MULTIPLY Num1 BY Num2 GIVING Result
```

```
END-IF
```

```
DISPLAY "Result is = ", Result
```

```
END-PERFORM.
```

```
STOP RUN.
```

DATA

Num1	Num2	Result	Operator
0	0	00	_

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Iteration-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE ZEROS.
01  Num2          PIC 9   VALUE ZEROS.
01  Result        PIC 99  VALUE ZEROS.
01  Operator      PIC X   VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    PERFORM 5 TIMES
```

```
        ACCEPT Num1
```

```
        ACCEPT Num2
```

```
        ACCEPT Operator
```

```
        IF Operator = "+" THEN
```

```
            ADD Num1, Num2 GIVING Result
```

```
        END-IF
```

```
        IF Operator = "*" THEN
```

```
            MULTIPLY Num1 BY Num2 GIVING Result
```

```
        END-IF
```

```
        DISPLAY "Result is = ", Result
```

```
    END-PERFORM.
```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
5	0	00	_

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Iteration-Program.  
AUTHOR. Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01 Num1 PIC 9 VALUE ZEROS.  
01 Num2 PIC 9 VALUE ZEROS.  
01 Result PIC 99 VALUE ZEROS.  
01 Operator PIC X VALUE SPACE.
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
PERFORM 5 TIMES
```

```
ACCEPT Num1
```

```
ACCEPT Num2
```

```
ACCEPT Operator
```

```
IF Operator = "+" THEN
```

```
ADD Num1, Num2 GIVING Result
```

```
END-IF
```

```
IF Operator = "*" THEN
```

```
MULTIPLY Num1 BY Num2 GIVING Result
```

```
END-IF
```

```
DISPLAY "Result is = ", Result
```

```
END-PERFORM.
```

```
STOP RUN.
```

DATA

Num1	Num2	Result	Operator
5	3	00	_

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```
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```
WORKING-STORAGE SECTION.
```

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01 Num1 PIC 9 VALUE ZEROS.  
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PROCEDURE DIVISION.
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PERFORM 5 TIMES
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IF Operator = "*" THEN
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MULTIPLY Num1 BY Num2 GIVING Result
```

```
END-IF
```

```
DISPLAY "Result is = ", Result
```

```
END-PERFORM.
```

```
STOP RUN.
```

DATA

Num1	Num2	Result	Operator
5	3	00	*


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```

```
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```

```
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```

```
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```

```
    STOP RUN.
```

DATA

Num1	Num2	Result	Operator
5	3	15	*

```
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WORKING-STORAGE SECTION.
```

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```
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```

DATA

Num1	Num2	Result	Operator
5	3	15	*

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```

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```

DATA

Num1	Num2	Result	Operator
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